Project Report(2025-26)

Fundamentals In AI & ML (CSA2001)

Title: Personalized Budget Tracker

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1. The Problem:

In an increasingly digital economy, many individuals—particularly students—lack a clear understanding of their spending patterns. Without a structured way to track expenses, users often face financial stress. Existing apps are often bloated with ads or require cloud subscriptions. There is a need for a lightweight, offline, and logic-driven tool that provides immediate financial feedback.

2. Why It Matters:

Financial literacy is a critical life skill. By building a tool that not only records data but analyzes it using **statistical volatility** and **logical rules**, we move from mere "data entry" to "actionable intelligence." This project reduces the gap between basic arithmetic and informed financial decision-making.

3. Approach to Solving It:

1. **Data Persistence:** I utilized Python's built-in `csv` module to ensure that user data is saved locally and remains accessible between program sessions.
2. **Statistical Analysis:** Rather than using libraries, I implemented manual algorithms to calculate the **Mean** and **Standard Deviation** of expenses to identify spending consistency.
3. **Logic Engine:** I developed a tiered feedback system using **First-Order Logic** principles to categorize the user's financial health (Healthy, Tight, or Critical).

4. Key Decisions:

No External Dependencies: I decided to avoid `pandas` and `matplotlib` to ensure the program is extremely lightweight and to demonstrate a deeper understanding of Python's standard library.

Standard Deviation as a Metric: I chose to include Standard Deviation to measure "Spending Volatility." A high value alerts the user that their spending is erratic, regardless of their total balance.

5. Challenges Faced:

Data Integrity: Handling empty files or corrupted rows in the CSV was a challenge. I solved this by implementing `try-except` blocks and a function that checks for the file's existence upon initialization.

Logic Mapping: Converting a budget "goal" into meaningful advice required careful calibration of thresholds (e.g., the 80% warning trigger).

6. What I Learned:

Through this project, I mastered the art of **File I/O** and learned how to translate mathematical formulas (like the variance formula) into clean, readable Python code. I also realized that powerful software doesn't always need complex libraries; solid logic is often more effective.